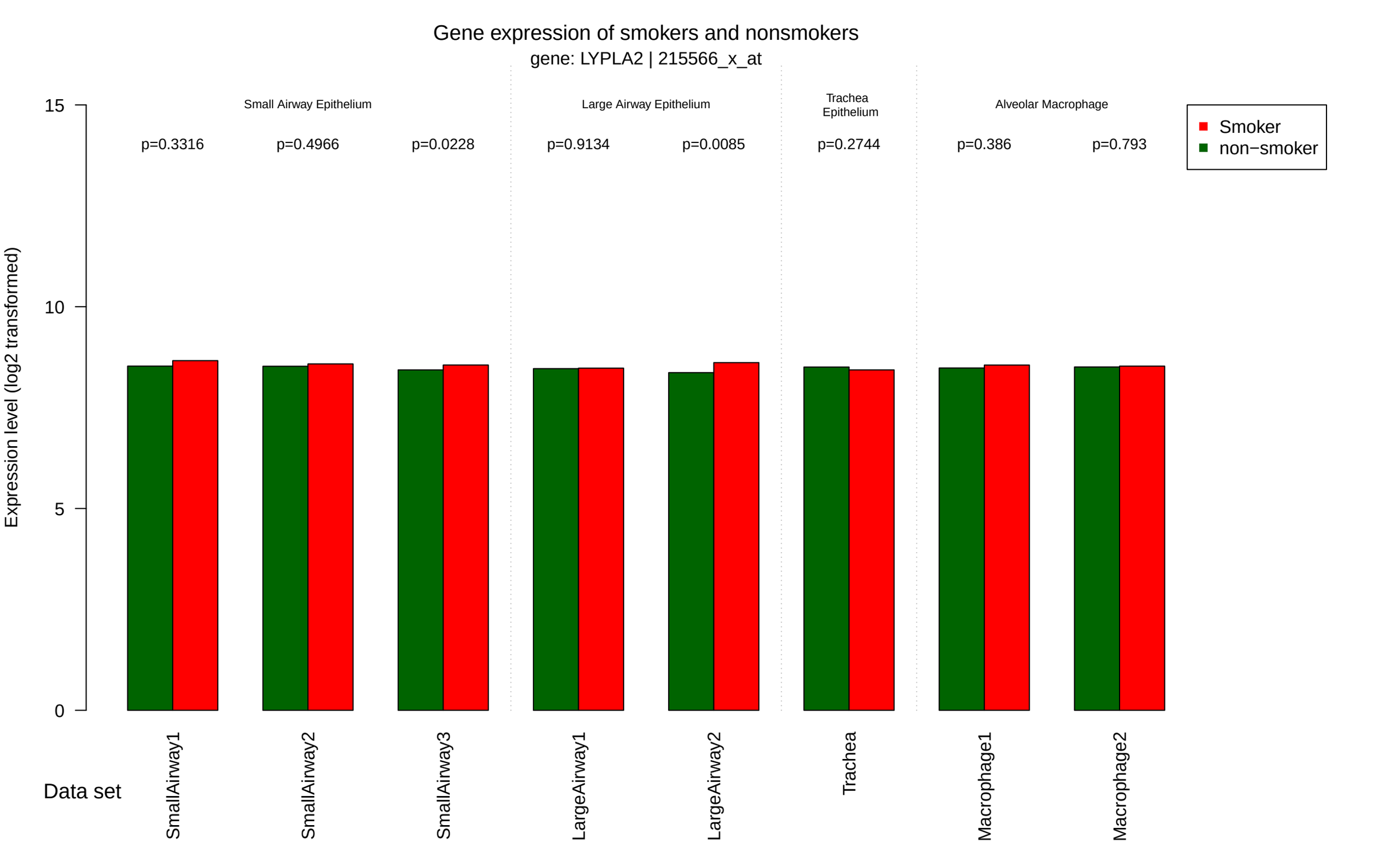
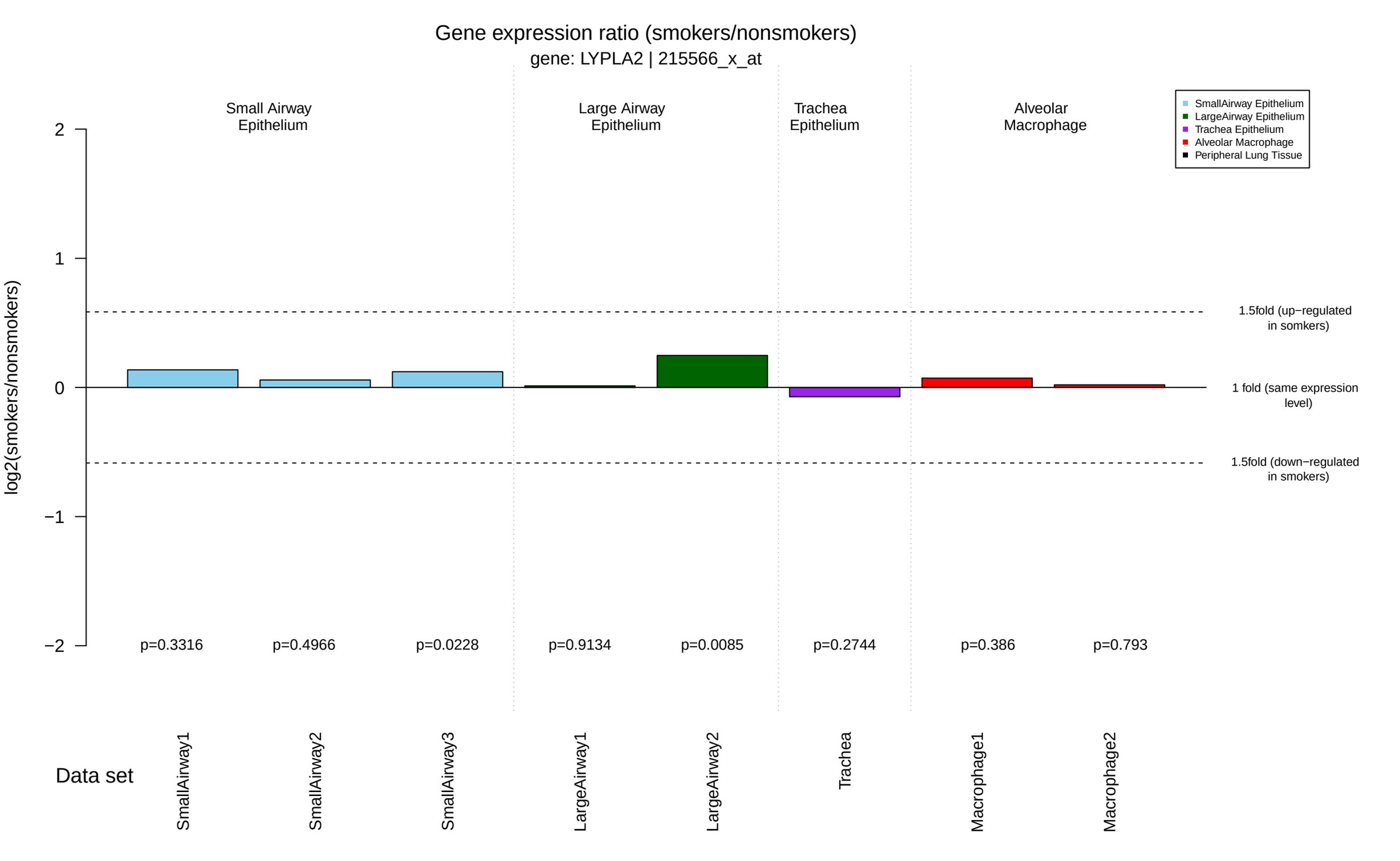
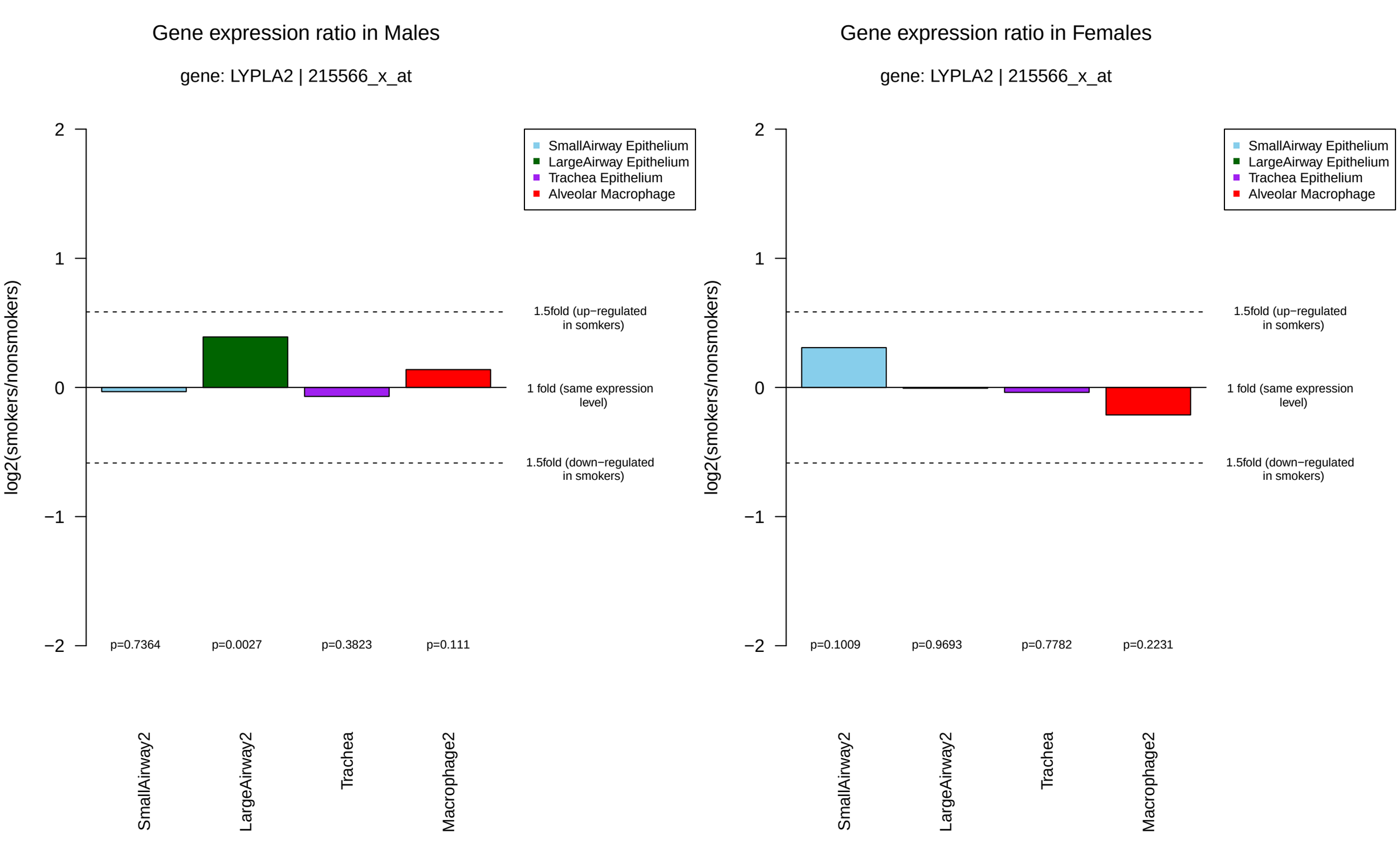
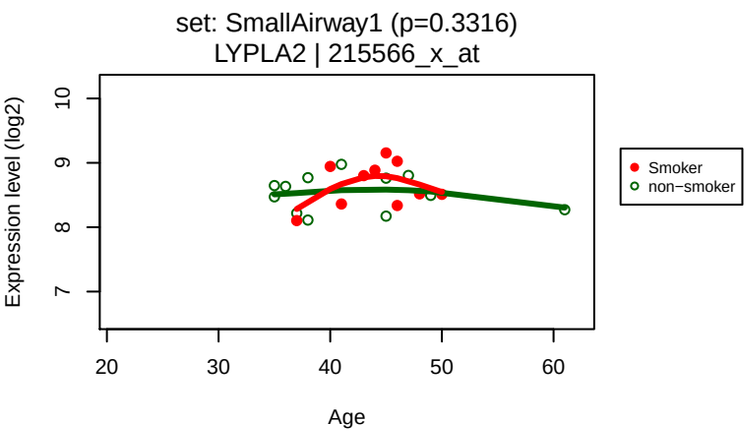
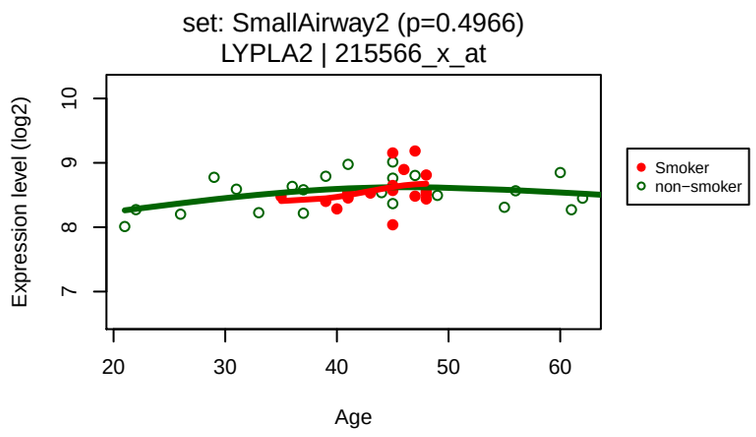
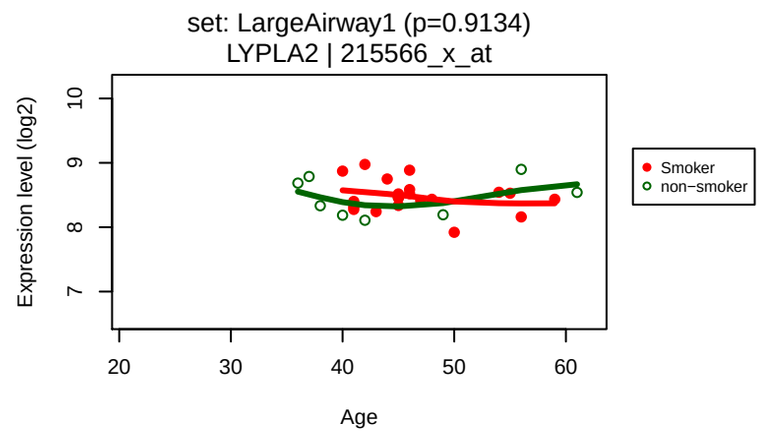

Gene expression of smokers and nonsmokers
gene: LYPLA2 | 215566_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3316
p=0.4966
p=0.0228
p=0.9134
p=0.0085
p=0.2744
p=0.386
p=0.793
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

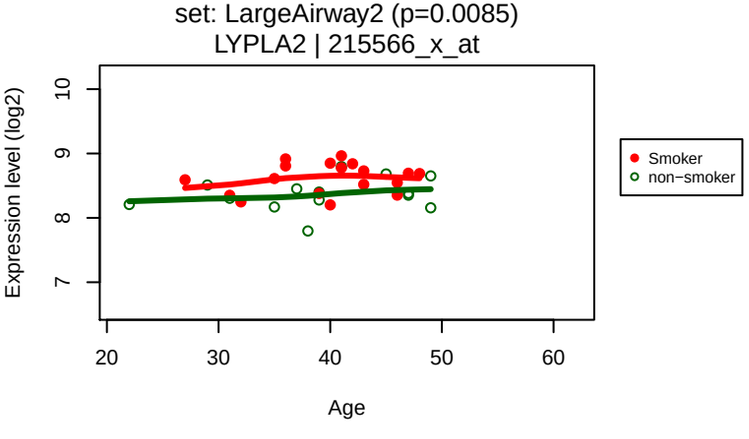

Gene expression ratio (smokers/nonsmokers)
gene: LYPLA2 | 215566_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3316
p=0.4966
p=0.0228
p=0.9134
p=0.0085
p=0.2744
p=0.386
p=0.793
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

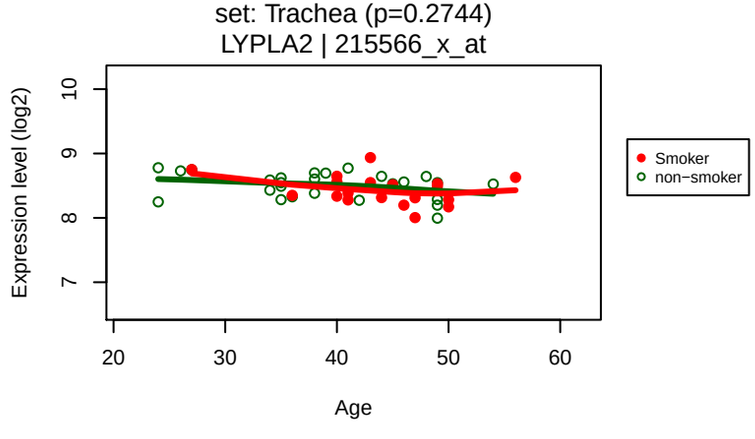

Gene expression ratio in Males
gene: LYPLA2 | 215566_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7364
p=0.0027
p=0.3823
p=0.111
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LYPLA2 | 215566_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1009
p=0.9693
p=0.7782
p=0.2231
SmallAirway2
LargeAirway2
Trachea
Macrophage2

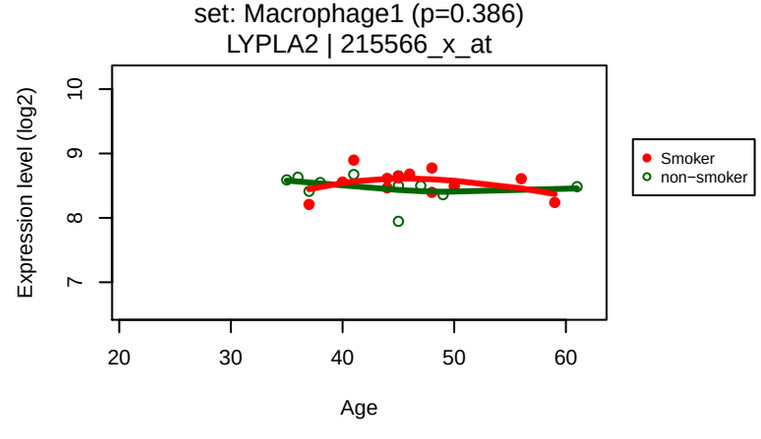

set: SmallAirway1 (p=0.3316)
LYPLA2 | 215566_x_at
Expression level (log2)
Age
Smoker
non−smoker

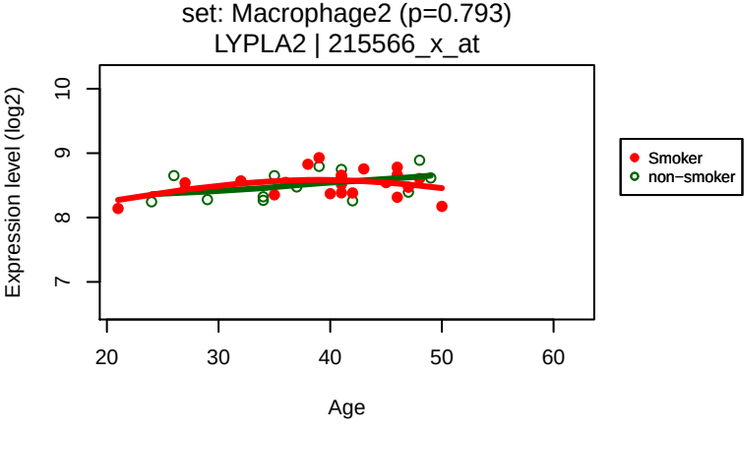

set: SmallAirway2 (p=0.4966)
LYPLA2 | 215566_x_at
Expression level (log2)
Age
Smoker
non−smoker

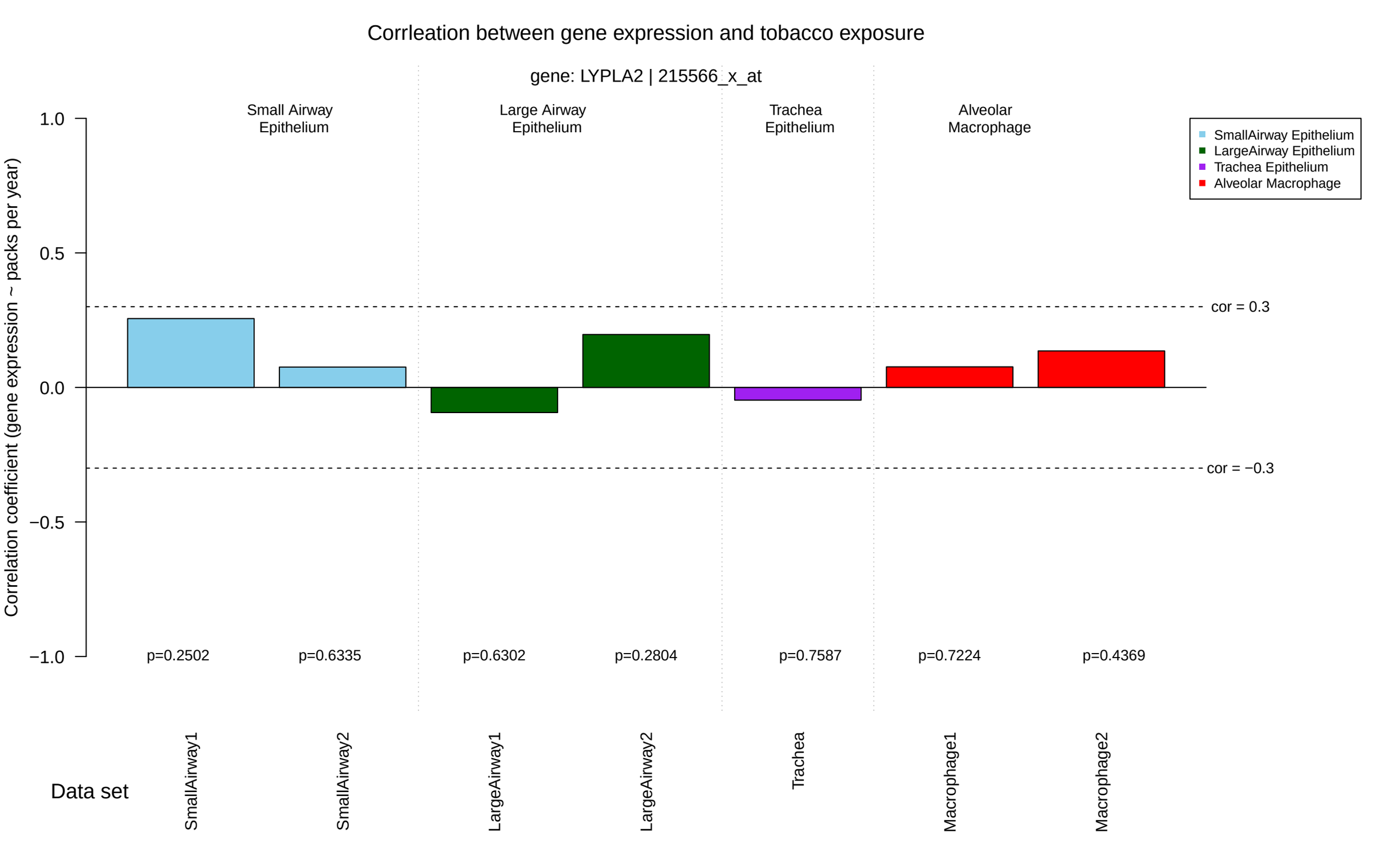

set: LargeAirway1 (p=0.9134)
LYPLA2 | 215566_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0085)
LYPLA2 | 215566_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2744)
LYPLA2 | 215566_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.386)
LYPLA2 | 215566_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.793)
LYPLA2 | 215566_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LYPLA2 | 215566_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2502
p=0.6335
p=0.6302
p=0.2804
p=0.7587
p=0.7224
p=0.4369
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2